

Assignment 1 — Univariate and Multivariate Statistics

FinTech 545 — Quantitative Risk Management

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Instructions

Be verbose. Explain clearly your reasoning, methods, and results in your written work. Write clear code that is well documented. With 99% certainty, you cannot write too many code comments.

Written answers are worth 8 points. Code is worth 2 points. 10 points total.

1. When finished, respond to the question in Canvas as “done.” We will record your grade there.
2. In your code repository, create a folder called “Assignment1.”
3. In that folder, include
 - a. a document (PDF) with your responses.
 - b. All code.
 - c. A README file with instructions for us to run your code.

Data for the problems is in the CSV files alongside this document in the class repository.

How These Problems Are Graded

Every problem below is in three parts, and the middle part is the one you are most likely to skip.

1. **Predict** – look at the data, or a plot of it, and say what you expect and why. Do this before you fit anything.
2. **Fit** – run the models.
3. **Reconcile** – say whether the fit agreed with you. If it did not, say what you had missed.

Most of the written credit sits on parts 1 and 3. Correct code with no reasoning attached does not earn full marks. A wrong prediction, honestly reported and then explained, earns nearly all of them. There are no points deducted for wrong predictions. We are grading whether you understand what the numbers mean, not whether you can call a library function.

1 Reading the Shape of a Sample (1.5 points)

problem1.csv holds a single series, x .

Predict. Compute the first four moments. From the skewness and the excess kurtosis alone, and before you fit anything, answer:

- a. Which of the distributions from Week 1 could plausibly have produced this sample, and which are ruled out? Name the moment that rules out each one you reject.

Fit and reconcile. Fit a Normal to the sample by matching the mean and variance. Then answer:

- b. How many observations fall below the Normal's 1% quantile? How many should there be? Report both numbers.
- c. If we used this fitted Normal as our model of the series, what would it get wrong? Say where in the distribution the error shows up, and in which direction.

2 A Regression Whose Errors Are Not Normal (2 points)

problem2.csv holds x and y .

Predict. Plot y against x before fitting anything.

- a. Describe what you see in the scatter. What distribution do you expect the error term to follow, and why?
- b. Ordinary least squares assumes normally distributed errors. Name which of the seven OLS assumptions this data violates, and predict what that violation will do – to the estimated slope, and separately to the standard error of that slope.

Fit. Estimate three models:

1. OLS, with the standard errors.
2. Maximum likelihood under a Normal error.
3. Maximum likelihood under a Student's t error.

Report α , β , and the fitted error parameters for each. Choose between the models using AICc.

Reconcile.

- c. Compare the three estimates of β . Did the violation move the slope as much as you predicted?
- d. The AICc prefers one model by a wide margin, yet the slopes are close together. What, specifically, is the rejected model getting wrong, if not the slope?
- e. Compare the 95% and the 99.5% quantiles of the fitted Normal error against those of the fitted t error. One is wider at 95% and the other is wider at 99.5%. Explain why, and say which one you would want if you were setting a capital buffer.

3 Pearson Against Spearman (1.5 points)

problem3.csv holds four series, x_1 through x_4 .

Predict. Plot every pair.

- Which pair do you expect the Pearson and Spearman coefficients to disagree on most? Which pairs do you expect them to agree on? Say why in each case.

Fit and reconcile.

- Compute both correlation matrices. Report the pair with the largest gap.
- Explain the mechanism that produces the gap. Which of the two coefficients is the honest description of that pair, and what question is the other one answering?

4 Conditional Distributions (1.5 points)

problem4.csv holds two correlated series, x_1 and x_2 .

Predict. Compute the sample covariance matrix, but nothing else yet.

- Write down the conditional variance of x_2 given x_1 using the partitioned result from Week 2, in the blocks of Σ . Say which of your series is which block. If you learn the value of x_1 , by what factor does your uncertainty about x_2 fall? Give that factor as a formula in the same blocks, and then as a number for this data.
- Under the multivariate normal, does that factor depend on the value of x_1 you observed? Point at the term in the formula that settles it, and answer before you look at the data.

Fit and reconcile.

- Write down the conditional mean of x_2 given x_1 , in the same blocks of Σ . Identify the coefficient on x_1 and say what it is in regression terms. Then plot the conditional expectation across the range of x_1 , with a 95% band around it, over a scatter of the data.
- Compute the fraction of observations that fall inside the band. Report it.
- Now split the sample by how far x_1 sits from its mean – inside one standard deviation, between one and two, and beyond two – and compute the coverage within each bucket. Report all three.
- The coverage is not flat across those buckets. Explain what that tells you about the data, and identify which assumption behind the constant width band has failed. Note carefully which parts of your answer to part (c) survive the failure and which do not.

5 Identifying an AR or MA Order (1.5 points)

problem5.csv holds a single series, x .

Predict. Plot the series, its ACF, and its PACF.

- a. Which of the two functions cuts off, and which decays? What order does that imply, and is it an AR or an MA process? Commit to a specific answer before fitting.
- b. State the rule you used. Cite the significance band you judged “cuts off” against.

Fit and reconcile.

- c. Fit AR(1), AR(2), AR(3), MA(1), MA(2), and MA(3). Report the AICc for each.
- d. Which model does the AICc select? Did it match your prediction?
- e. Compare the AR(2) and AR(3) fits. The third coefficient is small. Explain what AICc is trading off when it rejects the larger model, and why R^2 would not have made the same choice.

6 Code (2 points)

Your code is graded on whether we can run it and whether we can follow it. Include a README with the commands to reproduce every number in your write up. Comment the non-obvious steps, particularly anywhere you chose one convention over another.